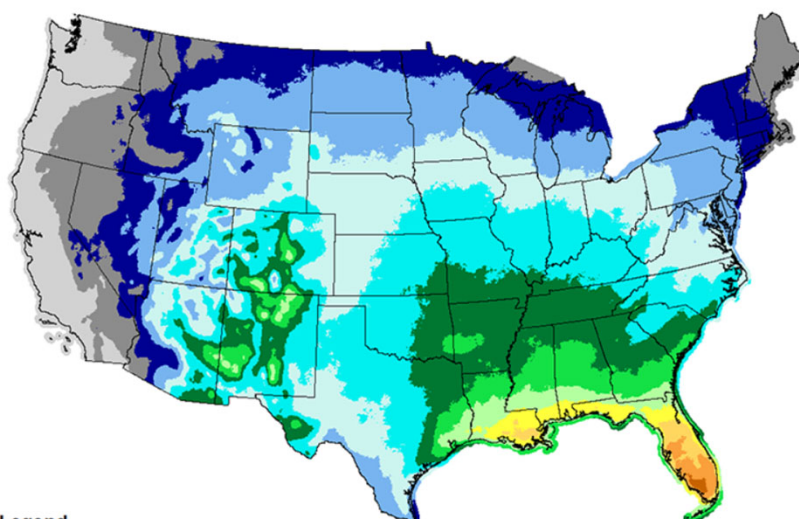




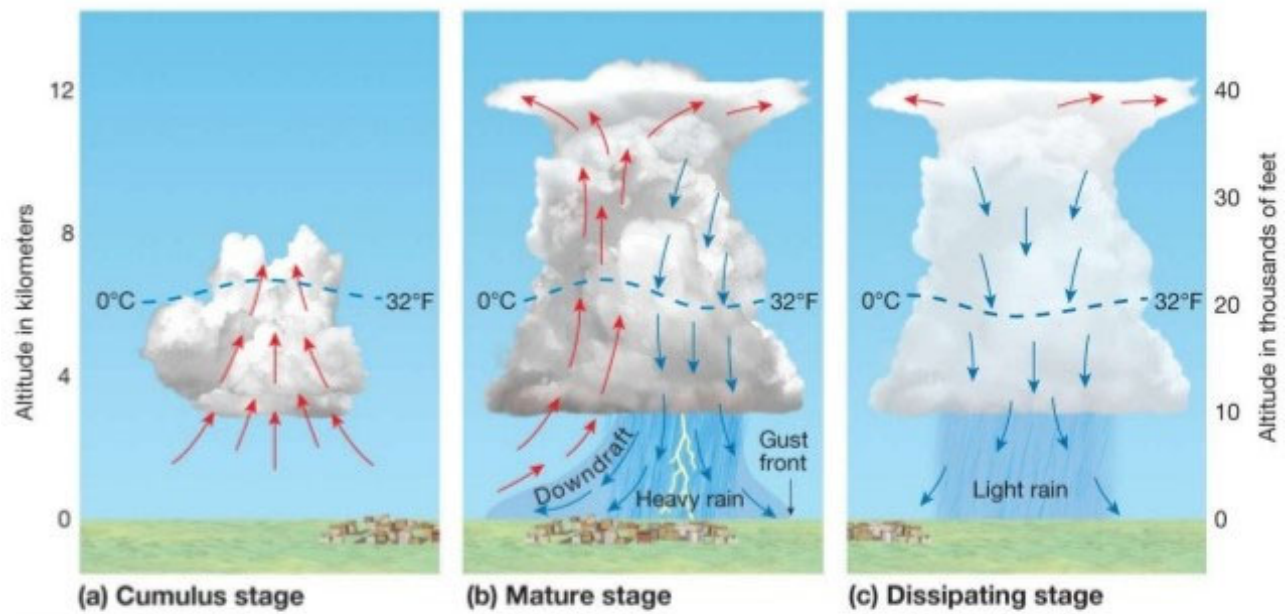
Annual Mean Thunderstorm Days (1993-2018)



Legend

Days	9+ to 18	27+ to 36	45+ to 54	63+ to 72	81+ to 90	99+ to 108
0+ to 9	18+ to 27	36+ to 45	54+ to 63	72+ to 81	90+ to 99	> 108

Annual number of thunderstorm days in the U.S. From: Koehler, Thomas L., 2019: Cloud-to-Ground Lightning Flash Density and Thunderstorm Day Distributions over the Contiguous United States Derived from NLDN Measurements: 1993-2018. Under review at Monthly Weather Review. Used by permission.

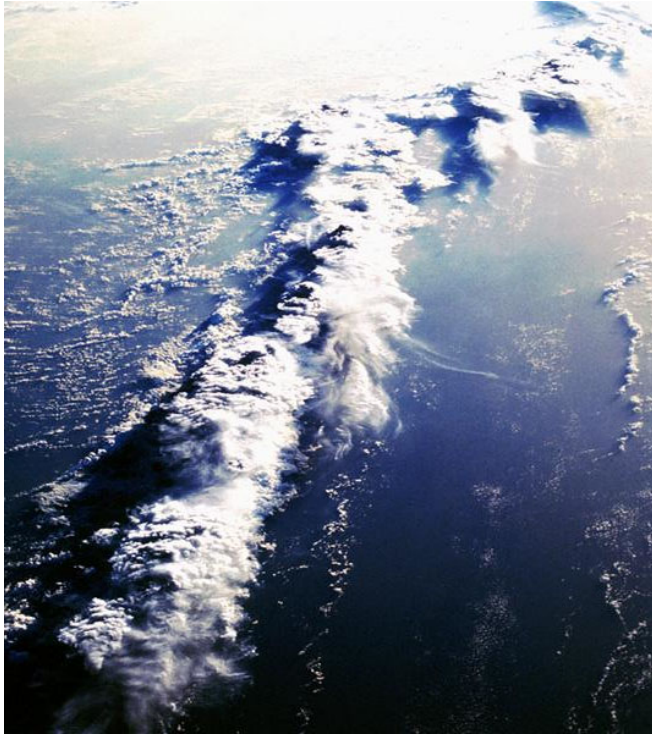




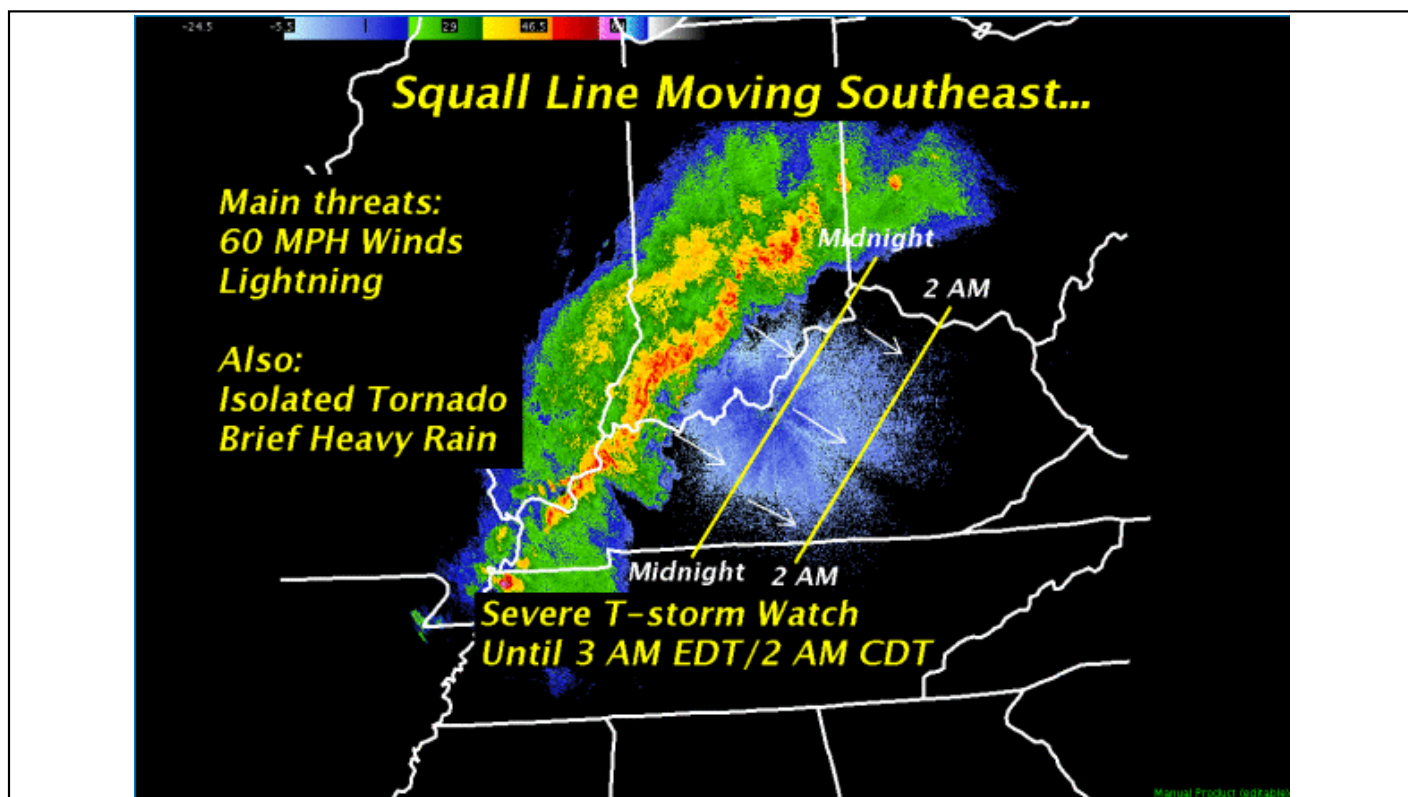
ON VIDEO

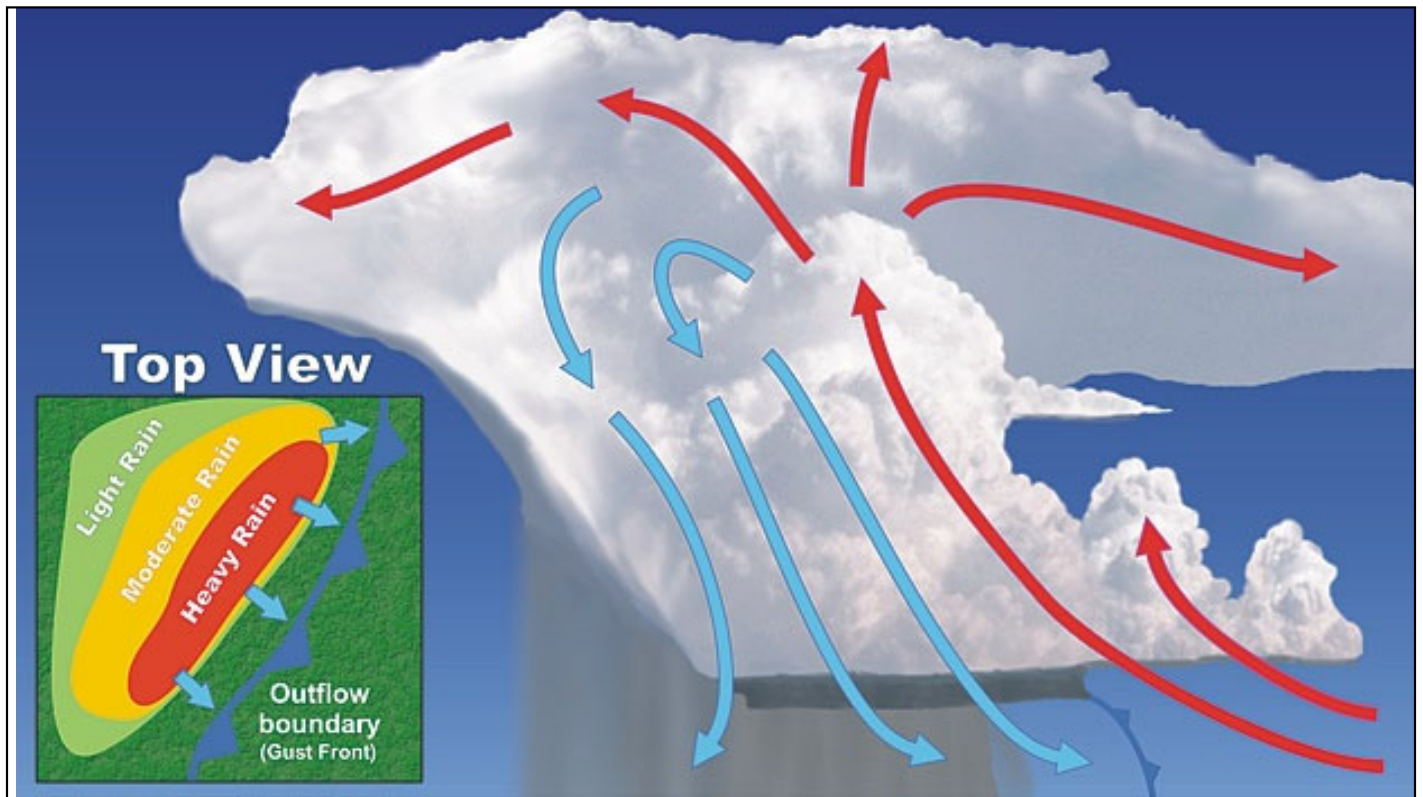
Nest

WXBRAD

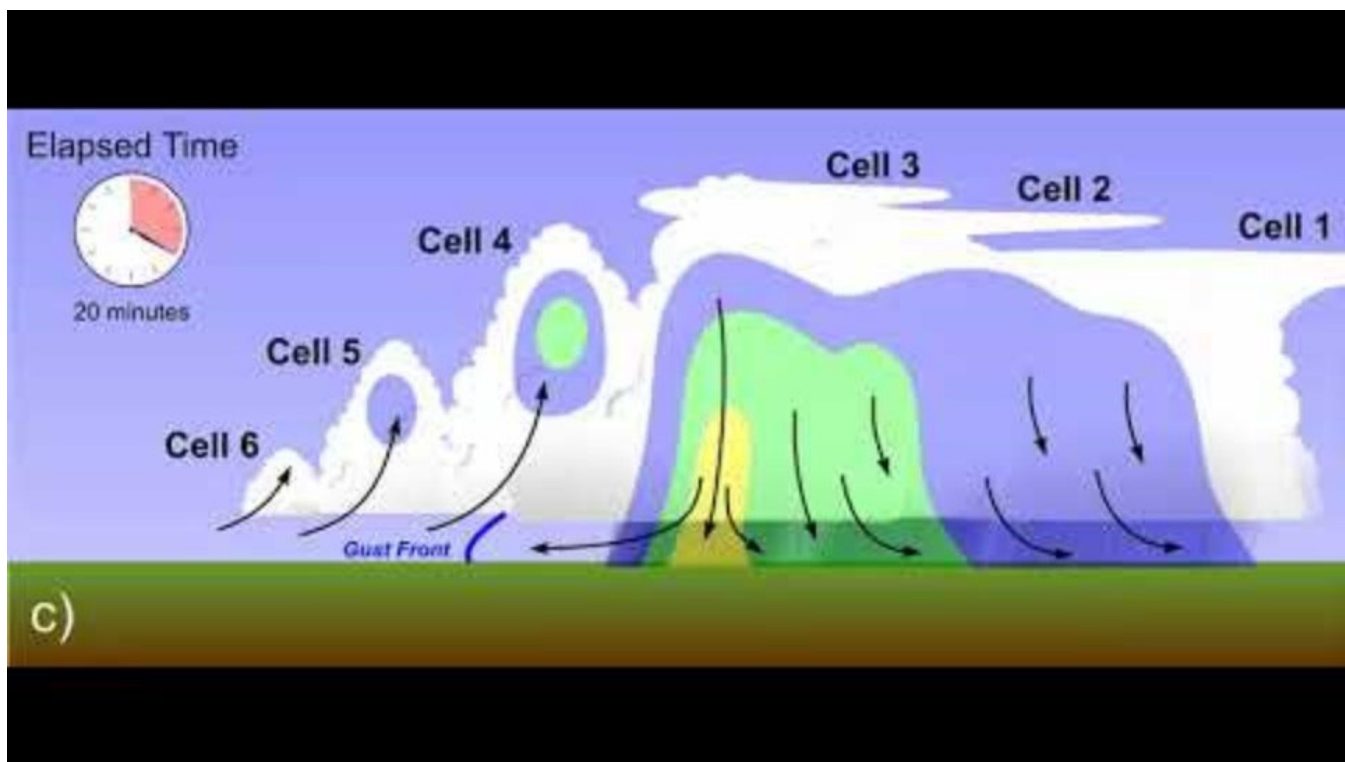


Self propagating





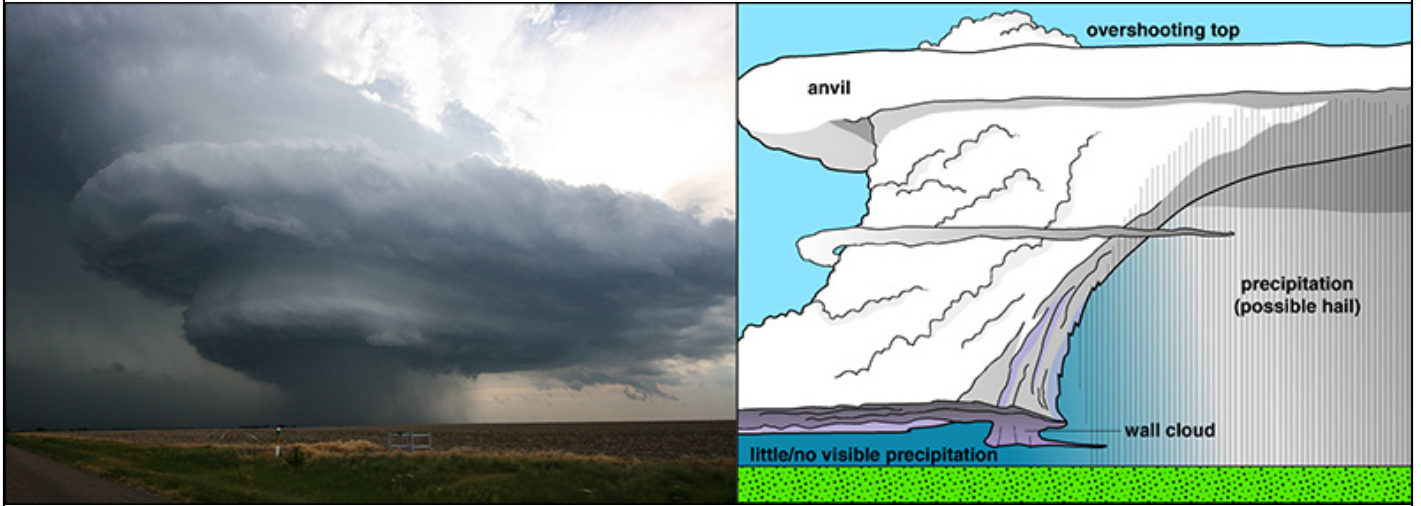
The outflow boundary is shown as a cold front because the air is often cool, having just descended from high in the atmosphere. These outflow boundaries can trigger new thunderstorms. If these thunderstorms move in a long linear line they can form a squall line thunderstorms. These are also called gust front thunderstorms.





Squall line severe thunderstorm



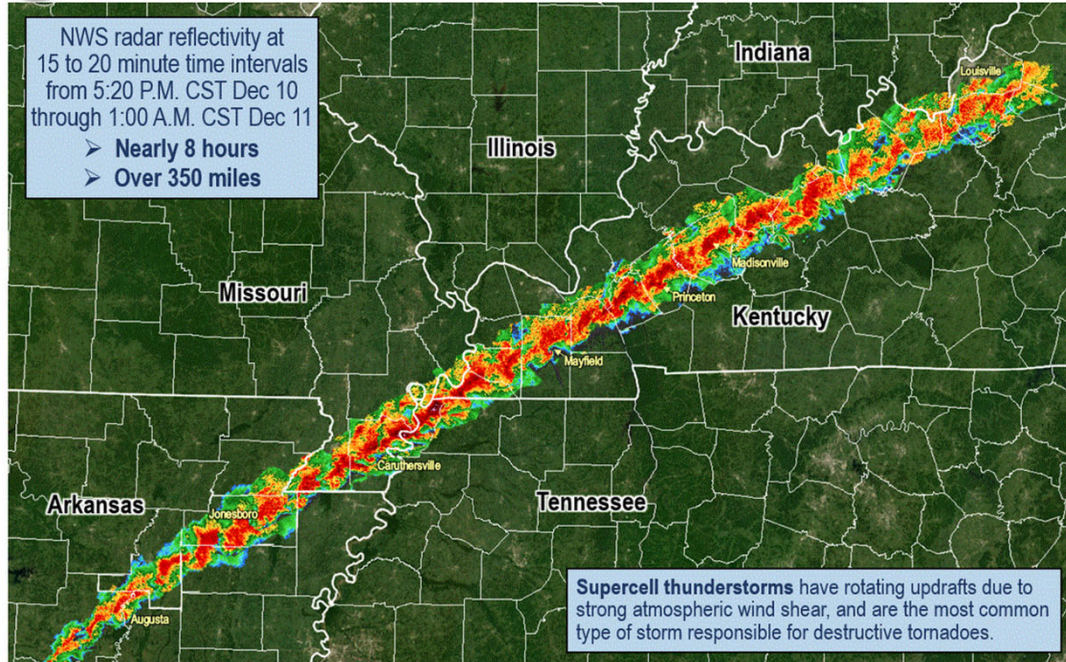






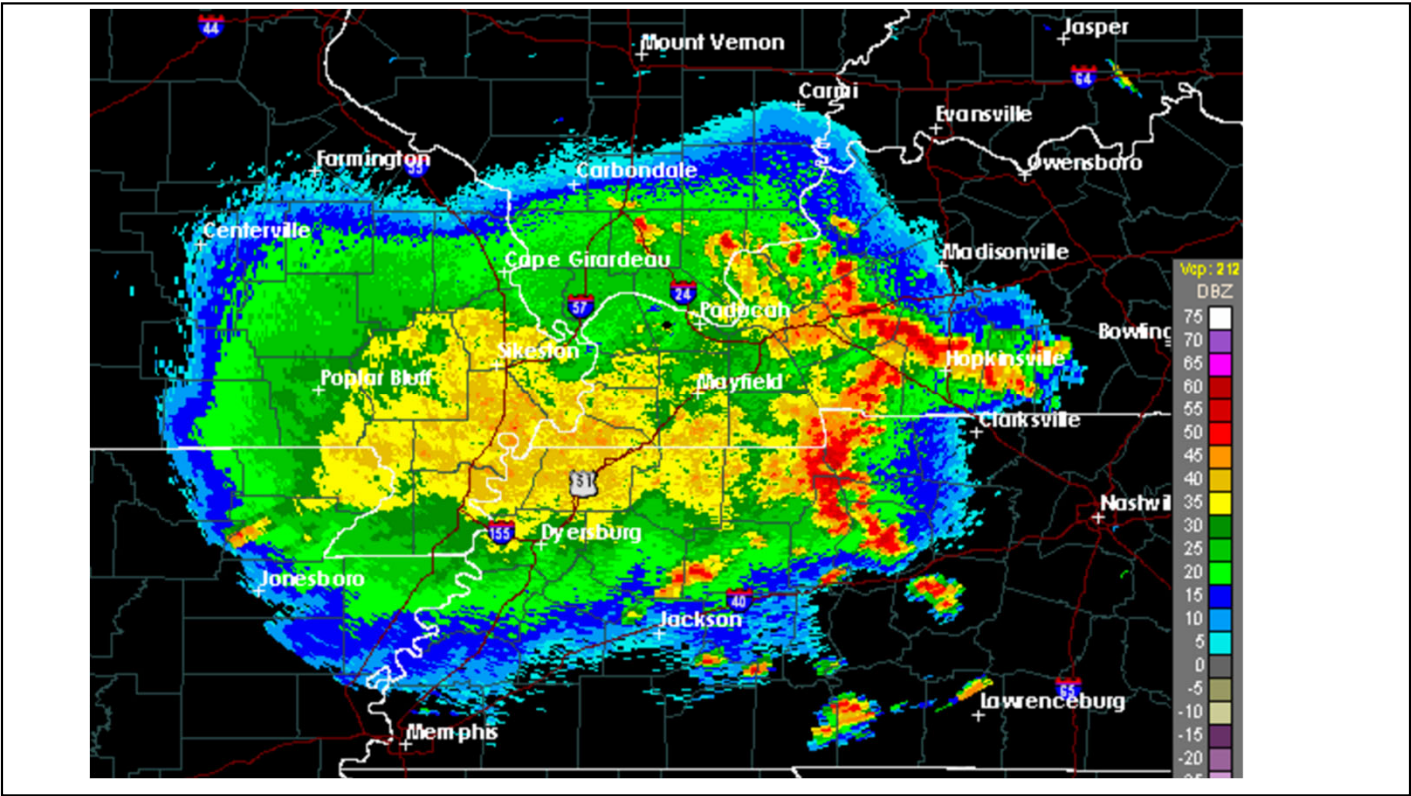


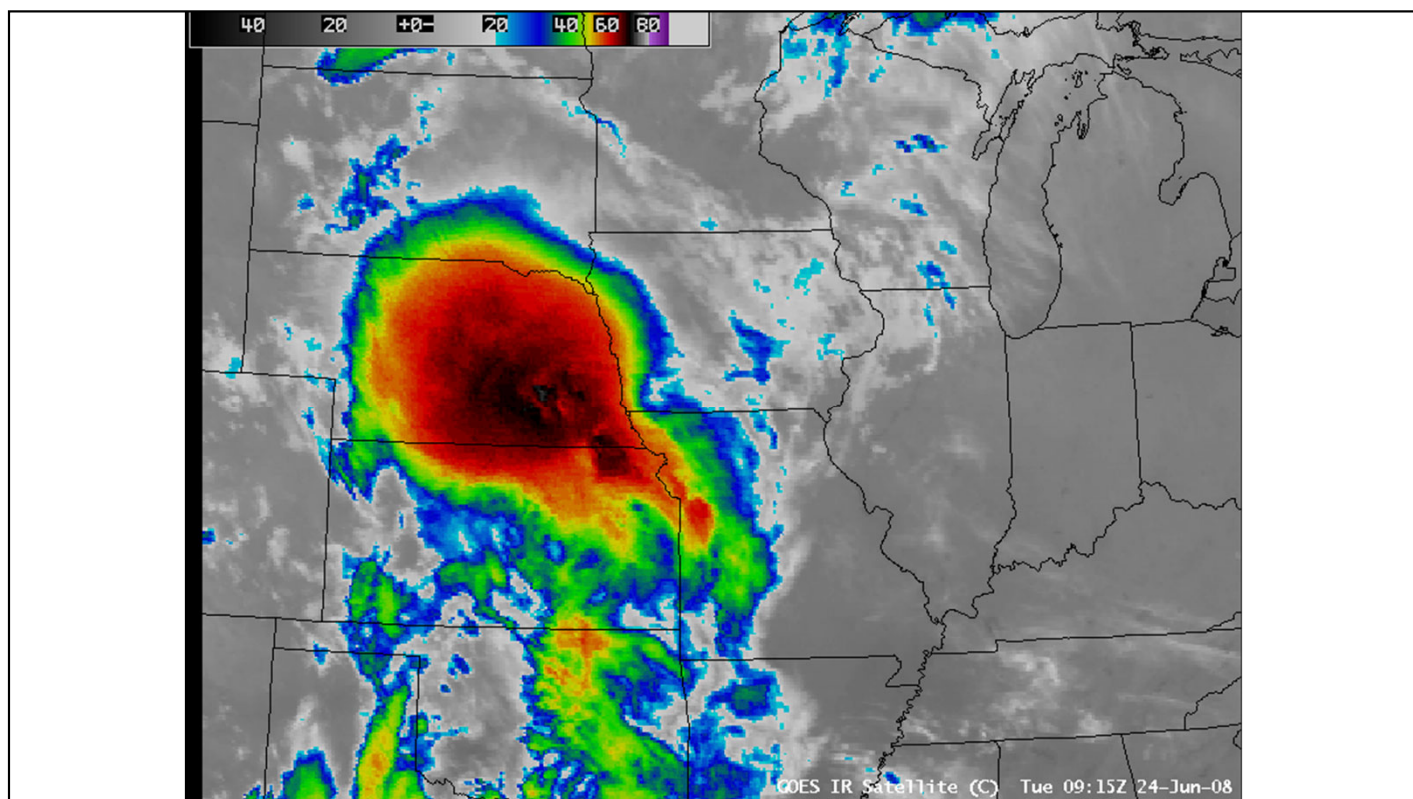
December 10-11, 2021: Long-lived Supercell Thunderstorm



National Weather Service

December 12, 2021





MCS

"THIS IS A PARTICULARLY DANGEROUS SITUATION..."

Have you ever noticed the wording "THIS IS A PARTICULARLY DANGEROUS SITUATION" in some of the severe weather watches issued by the NWS Storm Prediction Center?



This wording is used in **Tornado Watches** for rare situations when long-lived and intense tornadoes are likely.

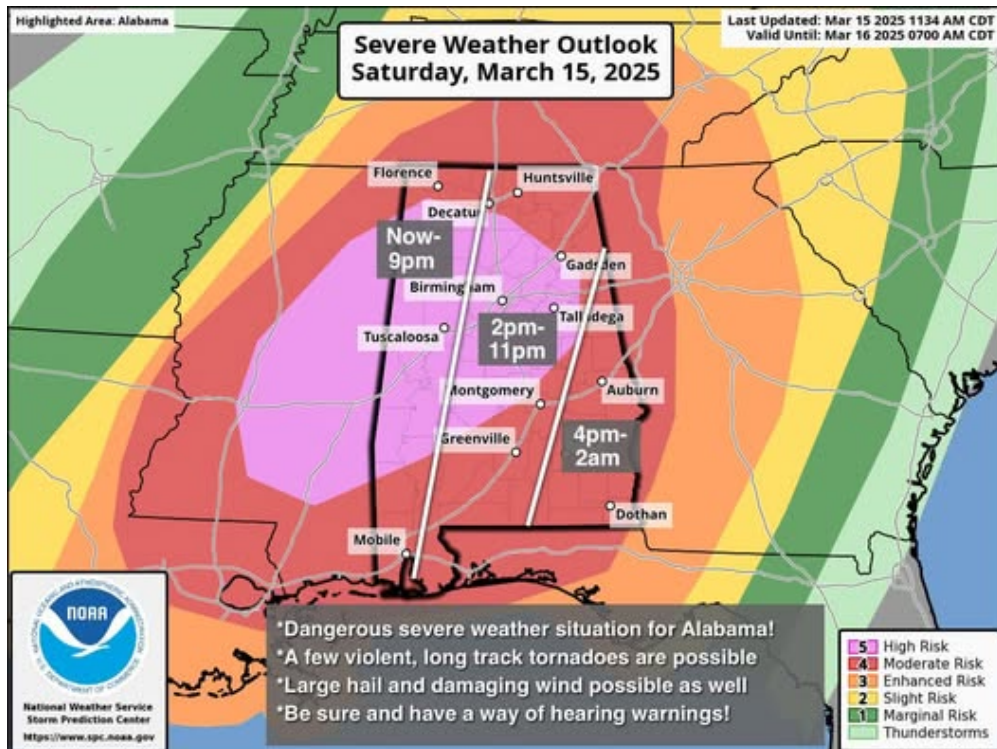
This enhanced wording may also accompany **Severe Thunderstorm Watches** for widespread and exceptionally intense straight-line wind events.

BE PREPARED:

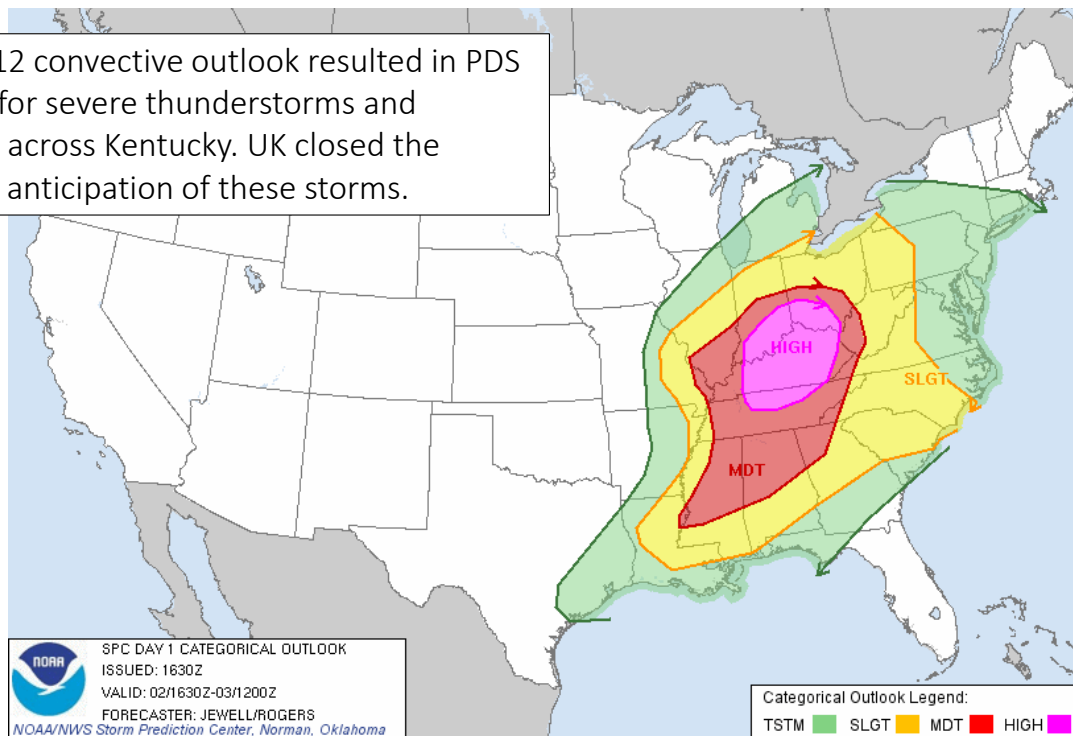


- Review your severe weather safety procedures
- Know the safest place to seek refuge ahead of time
- Consider canceling or postponing outdoor activities
- Have multiple ways to receive a warning
- Inform others of the heightened risk





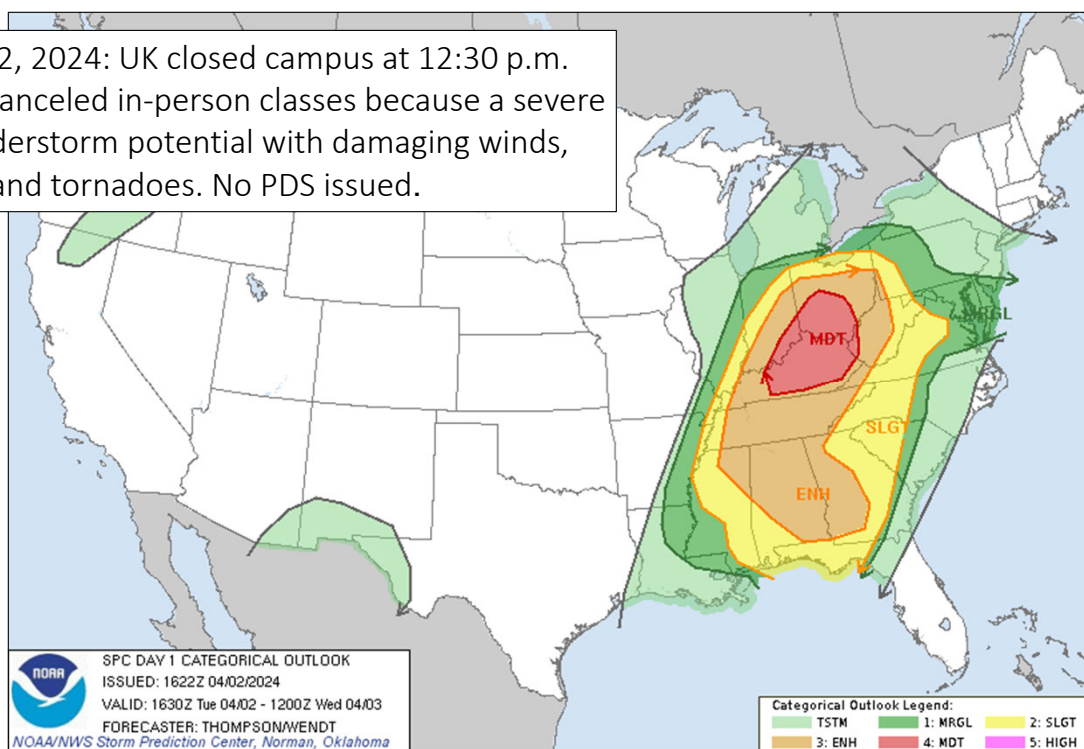
March 2012 convective outlook resulted in PDS issuances for severe thunderstorms and tornadoes across Kentucky. UK closed the campus in anticipation of these storms.



One of the most dangerous tornado days in Kentucky history.

2012 West Liberty KY tornado killed 2 people and injured 100. . The entire outbreak across the Midwest and Ohio Valley caused 41 deaths, with 22 fatalities in Kentucky, making it one of the deadliest tornado events in the state in decades.

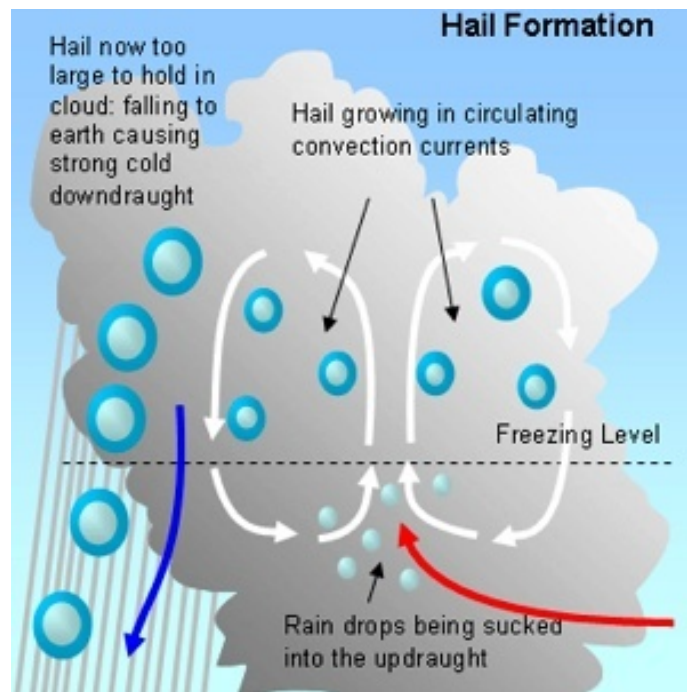
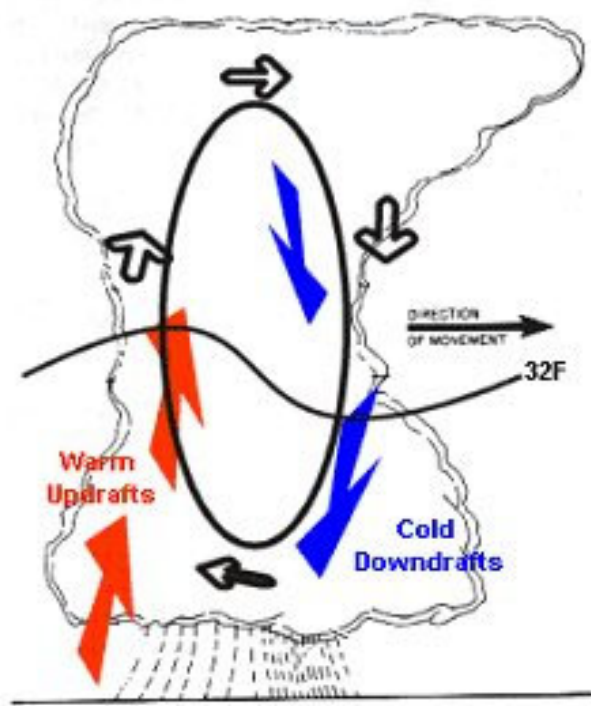
April 2, 2024: UK closed campus at 12:30 p.m. and canceled in-person classes because a severe thunderstorm potential with damaging winds, hail, and tornadoes. No PDS issued.



No PDS



Hailstones





Grapefruit-sized hailstone





Weather Related Commercial Airline Incidents in the U.S.

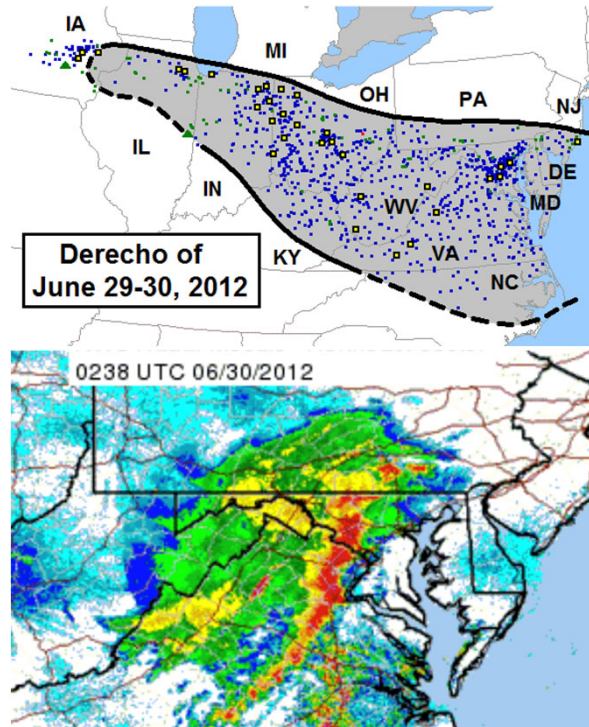
Flight	Date	Crash Location	Destination	Weather Hazard	Number of Fatalities
Capital Airlines 20	1/18/1960	Charles City, VA	Norfolk, VA	Aircraft Icing	50
Northwest Airlines 705	2/13/1963	Florida Everglades	Portland, OR	Thunderstorm (Turbulence)	43
Pan American World Airways 214	12/08/1963	Elkton, MD	Philadelphia, PA	Thunderstorm (Lightning)	81
Eastern Air Lines 66	6/24/1975	NYC	NYC	Thunderstorm (Microburst)	113
Southern Airways 242	1/04/1977	New Hope, GA	Atlanta, GA	Thunderstorm (Hail)	62
Air Florida 90	1/13/1982	Washington D.C.	Fort Lauderdale, FL	Aircraft Icing	70
Pan American World Airways 759	7/09/1982	Kenner, LA	Las Vegas, NV	Thunderstorm (Microburst)	153
Delta Air Lines 191	8/02/1985	Grapevine, TX	Los Angeles, CA	Thunderstorm (Microburst)	137
Continental 1713	11/15/1987	Denver, CO	Bosie, ID	Aircraft Icing	28
US Air 405	4/22/1992	NYC	Cleveland, OH	Aircraft Icing	27
US Air 1016	7/2/1994	Charlotte, NC	Charlotte, NC	Thunderstorm (Microburst)	68

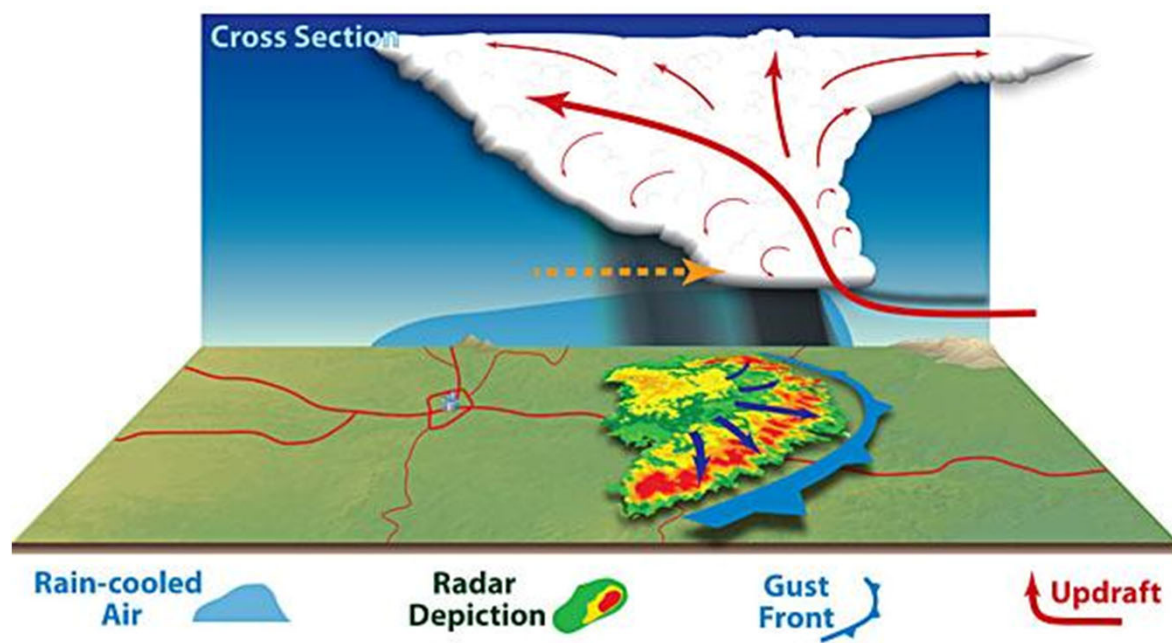
The Delta Air Lines Flight 191 airplane crash at Dallas/Fort Worth International Airport (DFW) occurred around 6:05 P.M. CDT on August 2, 1985. The National <https://www.weather.gov/fwd/delta191>

Transportation Safety Board's (NTSB) Accident Board determined that the cause of the incident was wind-shear associated with an intense thunderstorm downdraft that occurred at the north end of the airport along runway 17 Left (17L). Today we know this intense, localized downburst as a microburst, a weather phenomenon that was not well understood at the time of the accident. For information on an intense, but slightly larger downburst, see macroburst. The table below lists several aircraft incidents in which weather played a major role or was the direct cause for the loss of life and/or the loss of the airplane. The Delta Air Lines Flight 191 accident marked the 3rd event between 1975 and 1985 in which more than 100 fatalities occurred in an aircraft incident due to a microburst in the United States. The events on August 2, 1985, as well as other dates (see table below) likely helped to drive the need to reform aviation weather safety. These needs included the improved the detection of hazardous weather to aircraft using remote sensing instruments, equipping pilots and aircraft with tools to identify weather hazards to aircraft and continued collaboration between the National Weather Service (NWS) and Federal Aviation Administration (FAA).



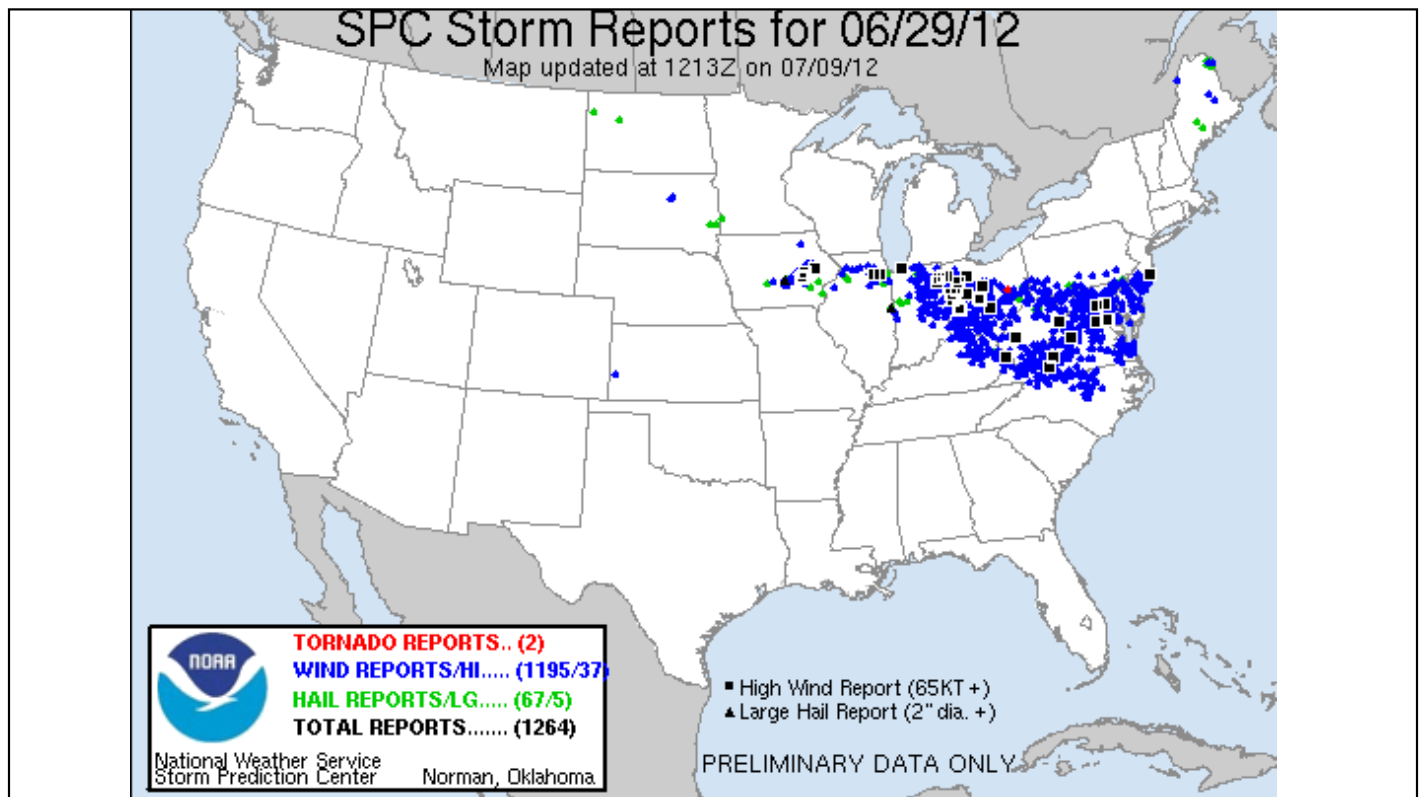
https://www.youtube.com/watch?v=a_G2KRzha7o





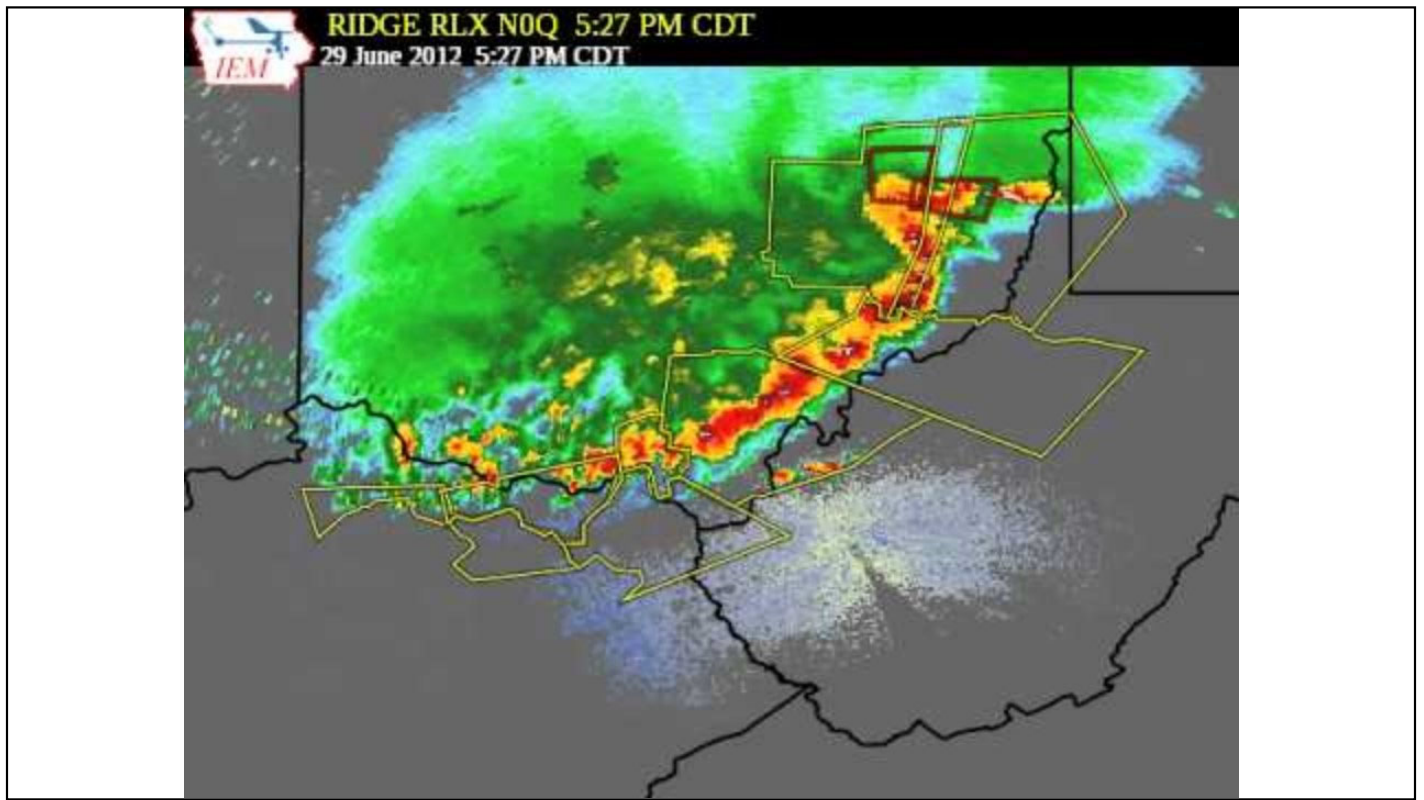


Approaching gust front and derecho in Chicago in June 2012



June 29, 2012 Eastern US- Mid Atlantic Derecho

<https://earthsky.org/earth/videos-and-images-violent-us-storm-of-june-29-2012/>



<https://youtu.be/0MG7aOjZciY?si=tEMTTTPPxwRWrHN7D>

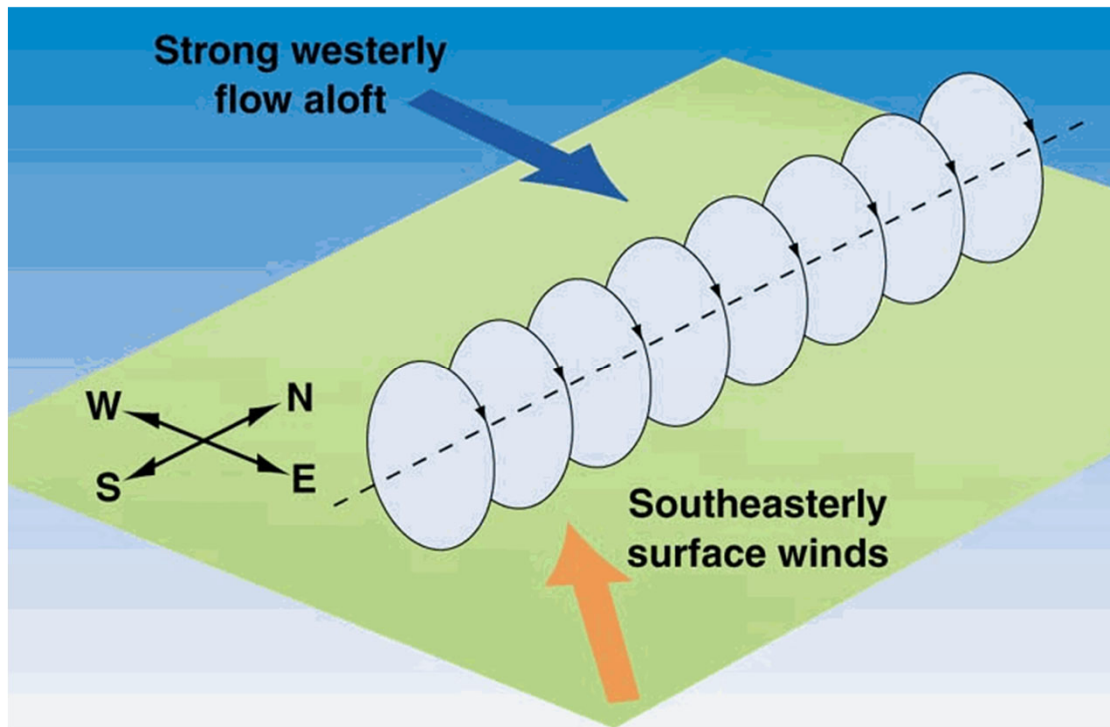


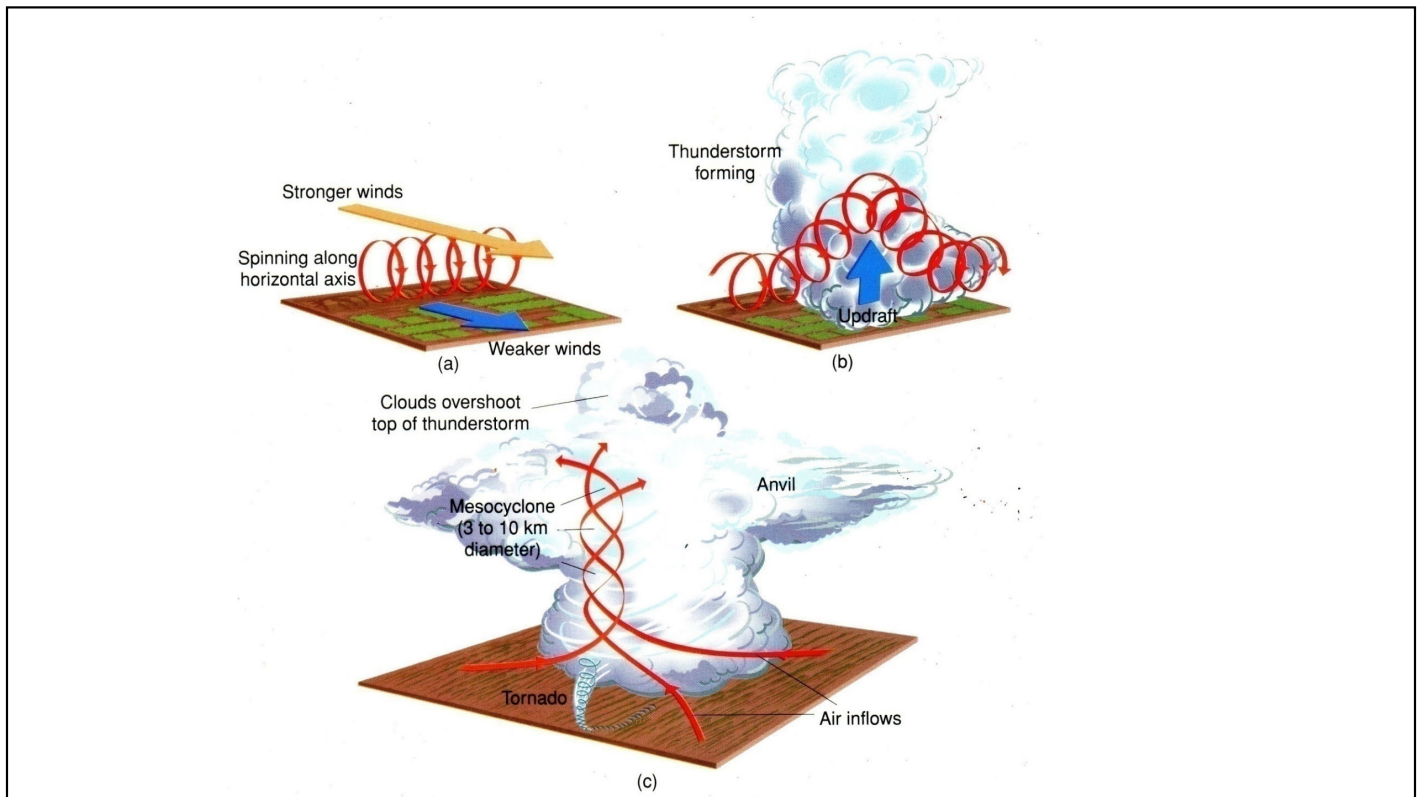
<https://youtu.be/ZtauyyeF-mo?si=pNkOuGxYSJFBqrqt>



<https://youtu.be/2GuCguODZbY?si=l7-IBVDDnsGX3beR>





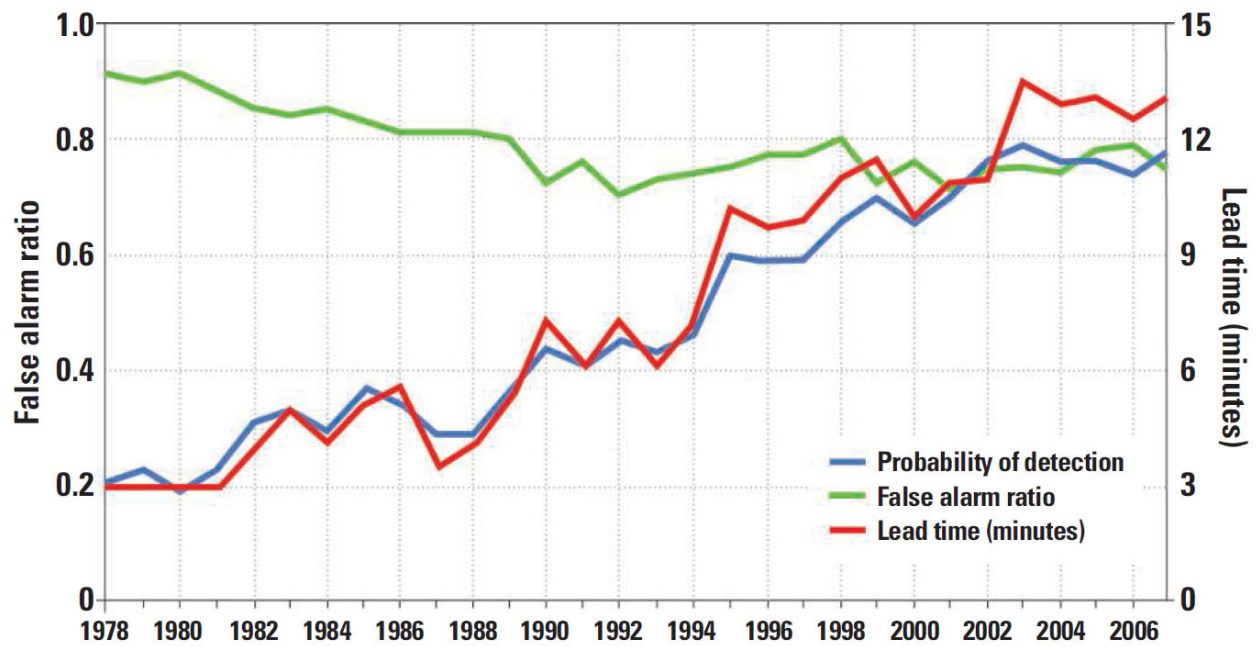


Keep in mind that the mesocyclonic circulation of a supercell is not the tornado



<http://www.accuweather.com/en/weather-news/indiana-stage-collapse-gustnad/53798?partner=>

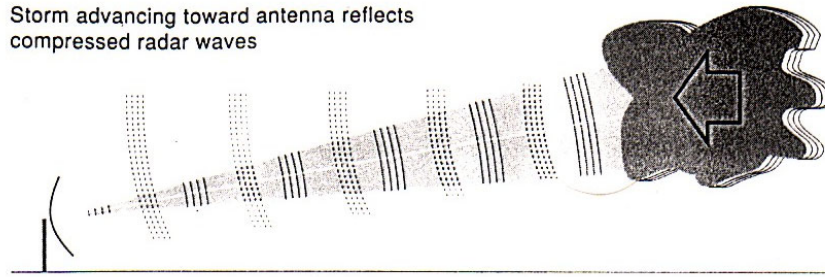




The false alarm ratio has held steady the last twenty years may be the result of using radar detection to issue tornado warnings.



Storm advancing toward antenna reflects compressed radar waves



Storm moving away from antenna reflects expanded radar waves

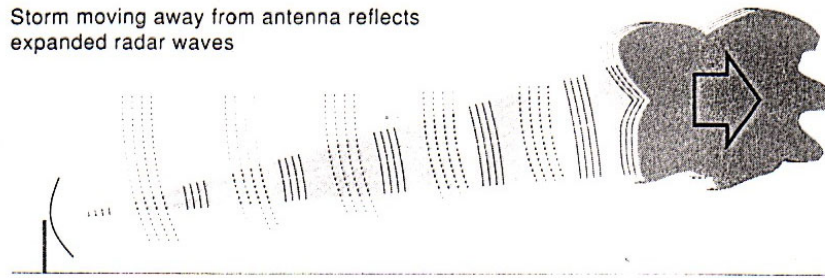
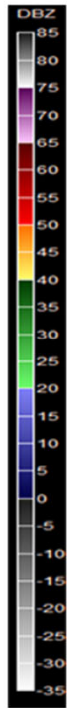


Fig. 8.4. Doppler radar computes a storm's velocity from the compression or expansion of radar waves reflected by the precipitation-size droplets of an advancing or retreating storm.

Based on Doppler effect: an oncoming train will have a higher whistle pitch as sound waves are compressed by the forward motion of the train

Tracks the motion of clouds by pulsing atmosphere with microwave radiation and then measuring shift in wavelength to calculate the cloud's direction or motion and velocity

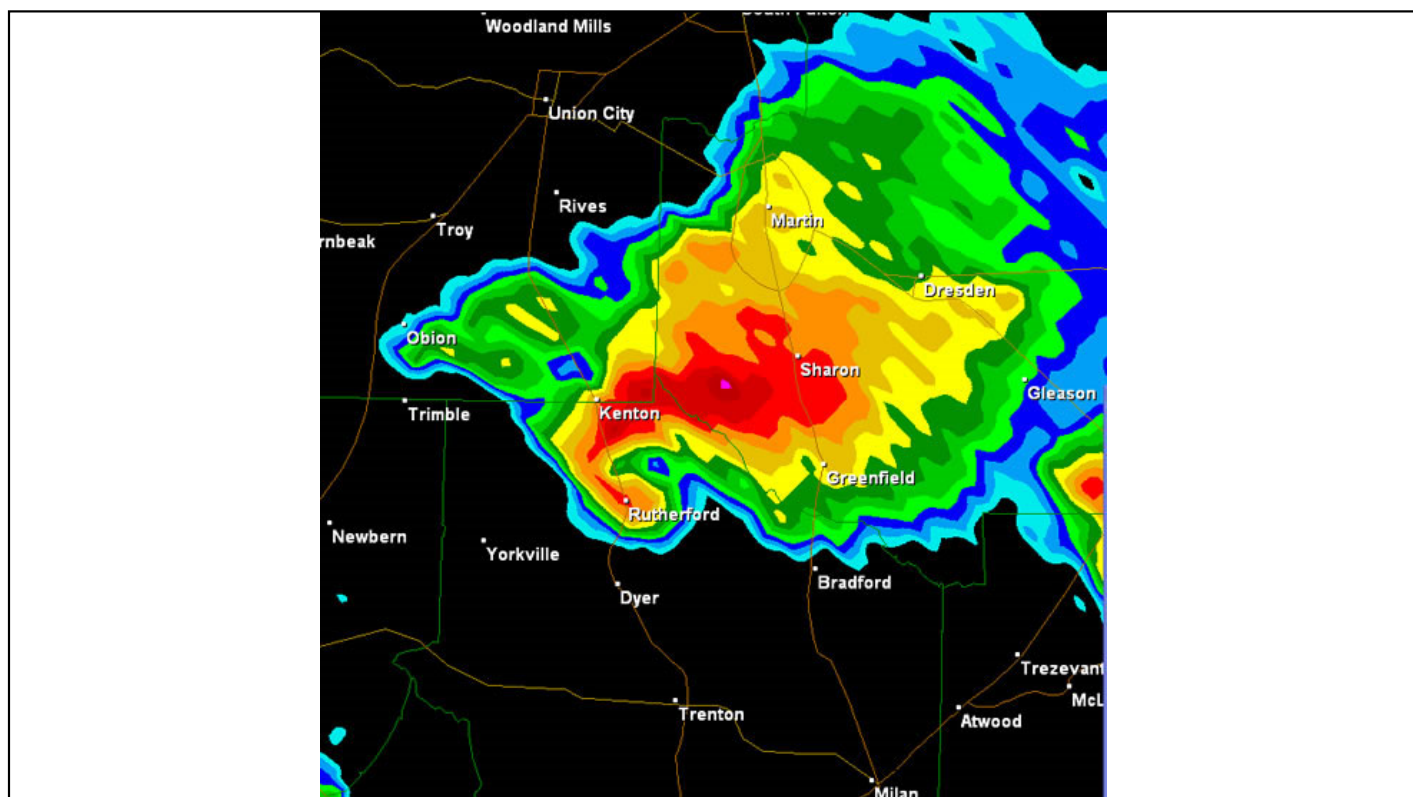




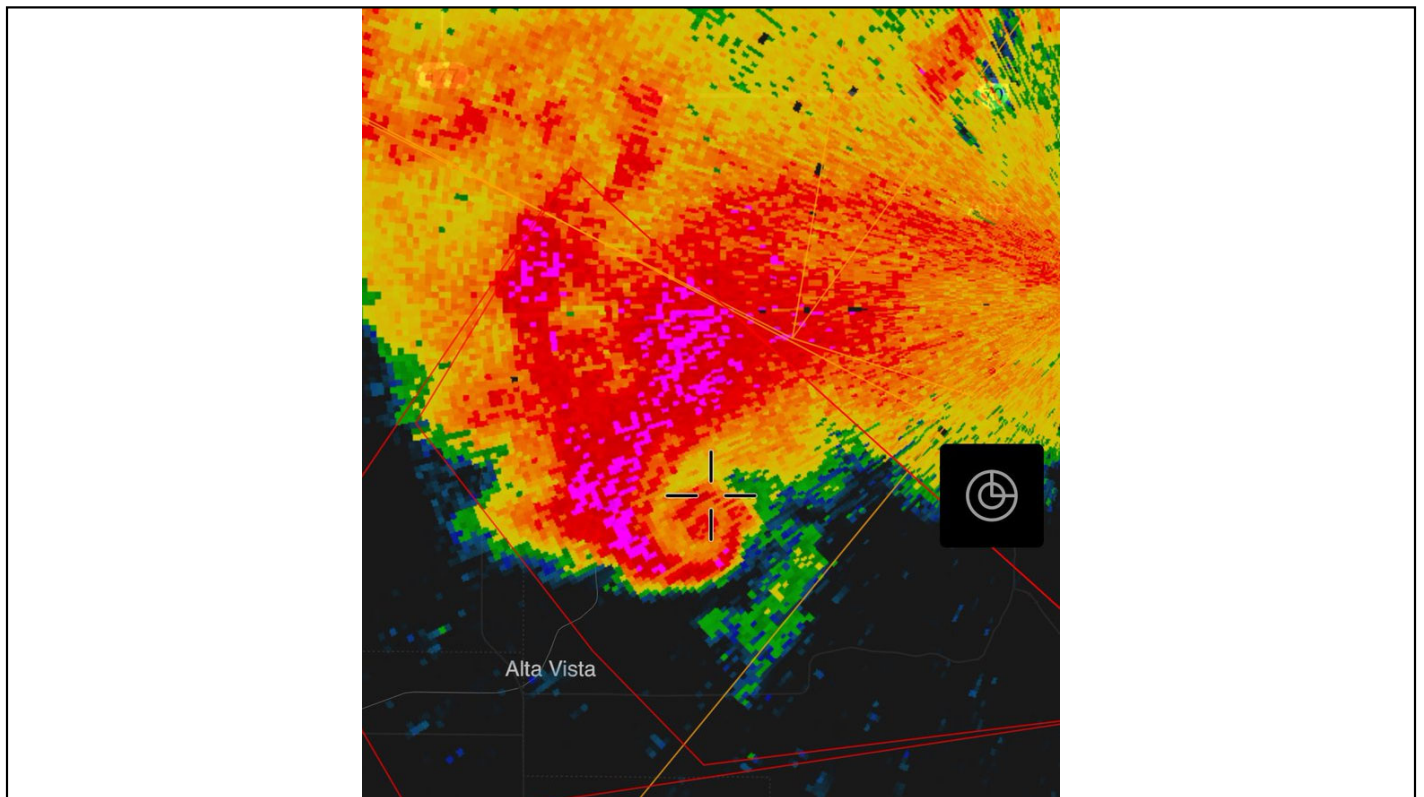
General Reflectivity Guidelines

dBZ	Expected Intensity
-35 to 0 dBZ	Extremely light (drizzle/snow)
0 to 20 dBZ	Very light precipitation or general clutter
20 to 40 dBZ	Light precipitation*
40 to 50 dBZ	Moderate precipitation*
50 to 65 dBZ	Heavy precipitation or some hail
> 65 dBZ	Extremely heavy precipitation including water-coated hail

<https://www.noaa.gov/jetstream/reflectivity>



Doppler reflectivity showing supercell with mesocyclonic rotation and potential tornado



Exceptionally well-formed Doppler radar reflectivity image of tornado near Alta Vista Kansas, March 13 2024

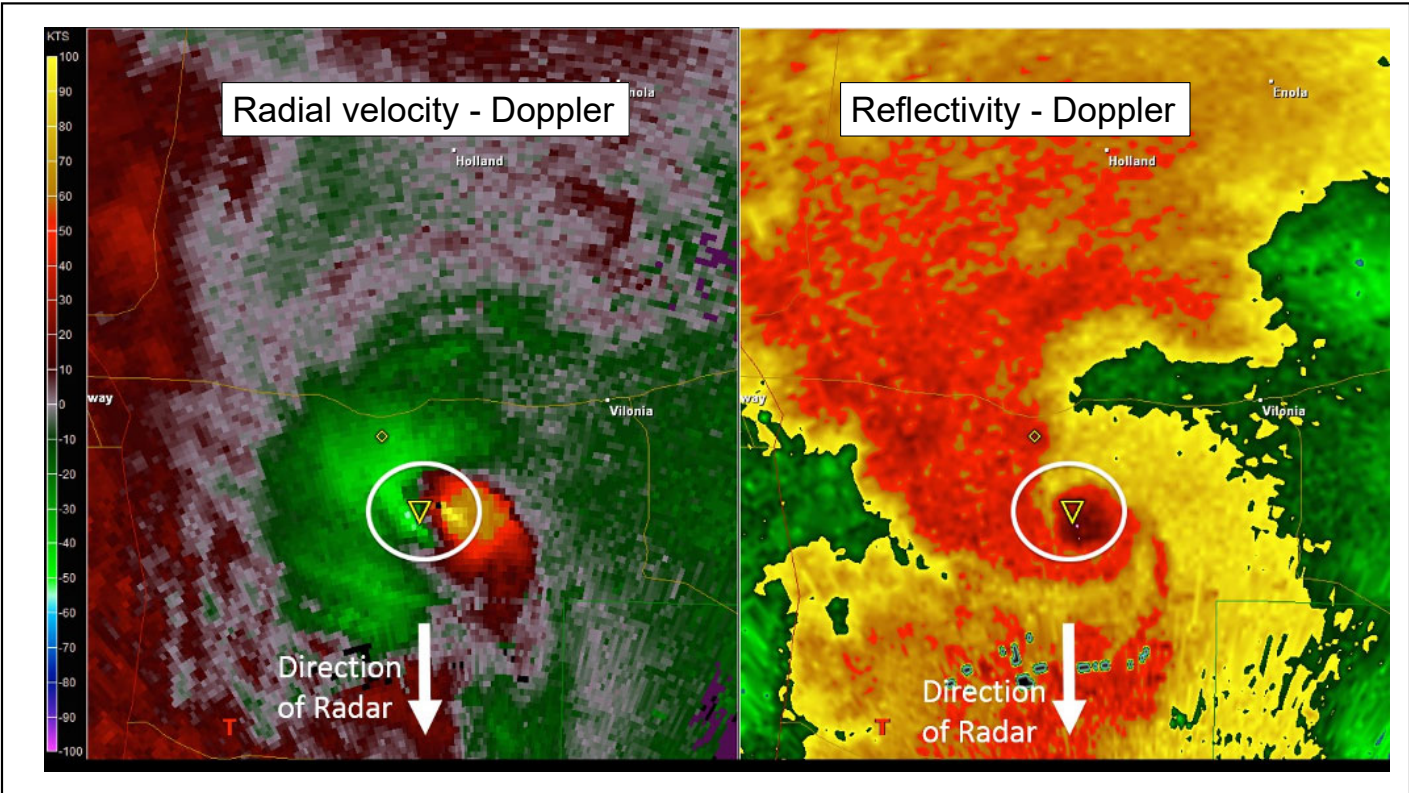
<https://www.washingtonpost.com/weather/2024/03/14/gorilla-hail-storm-kansas-missouri/>

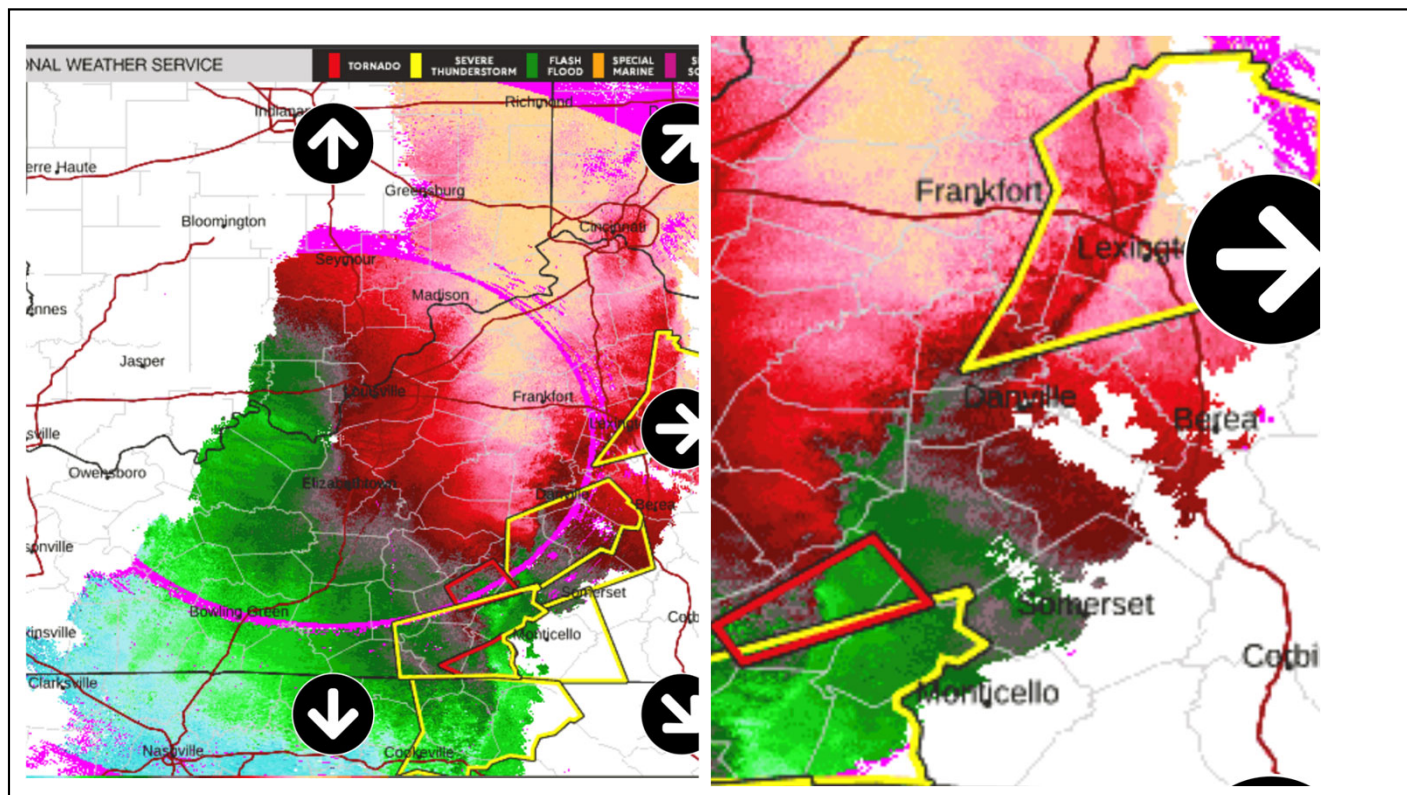


<https://youtu.be/lHBZylcxIvw?si=Bmd0qTYXFVeuu34z>

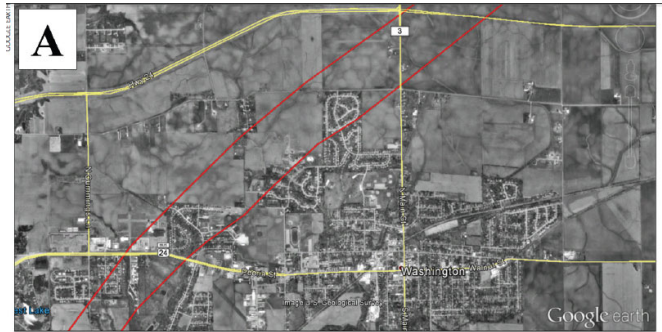


Doppler radar in this image is to the northeast. The red colors indicate winds that are moving away from the radar. The green indicate winds that are moving toward the radar. This sharp change in wind direction is indicative of mesocyclonic rotation and potentially a tornado





March 16, 1:30 am



Washington, IL in A) April 6, 1994 and B) November 18, 2013 with the November 17, 2013 tornado footprint overlaid as an example of the expanding bull's-eye effect. A majority of the development struck by the tornado path was built within the last 10 years.

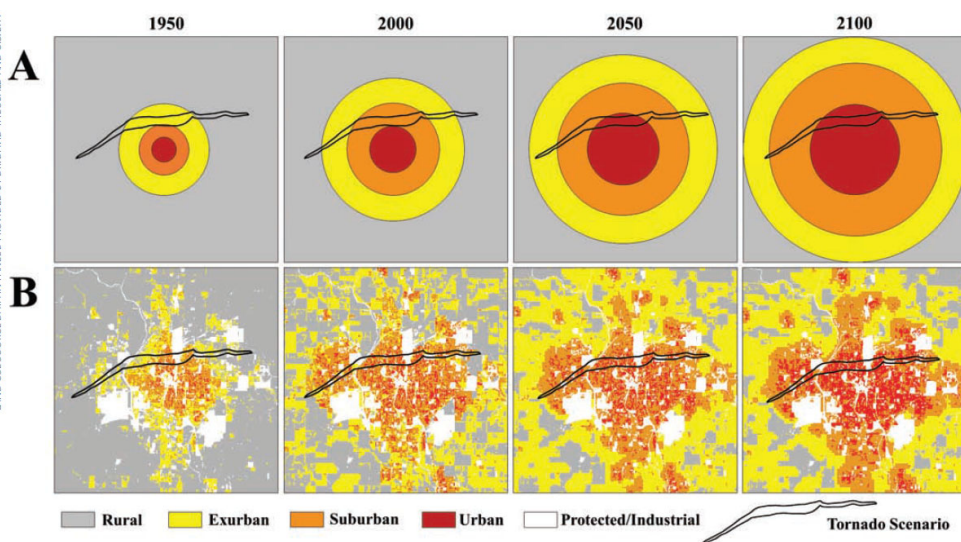
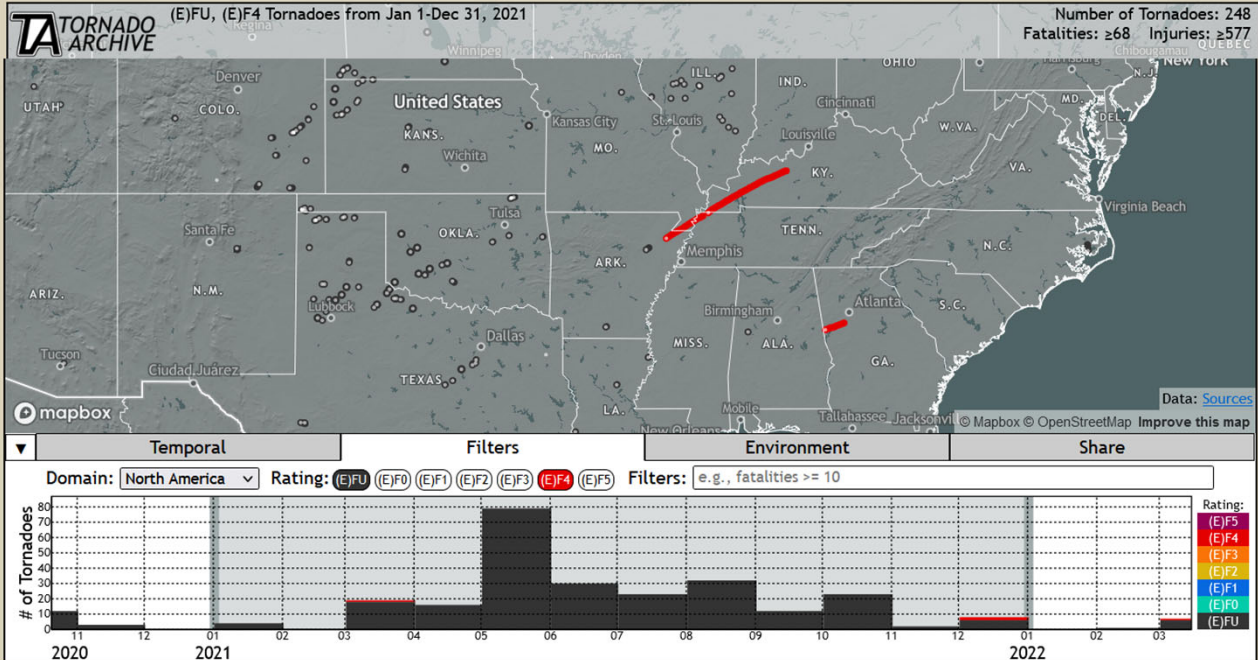
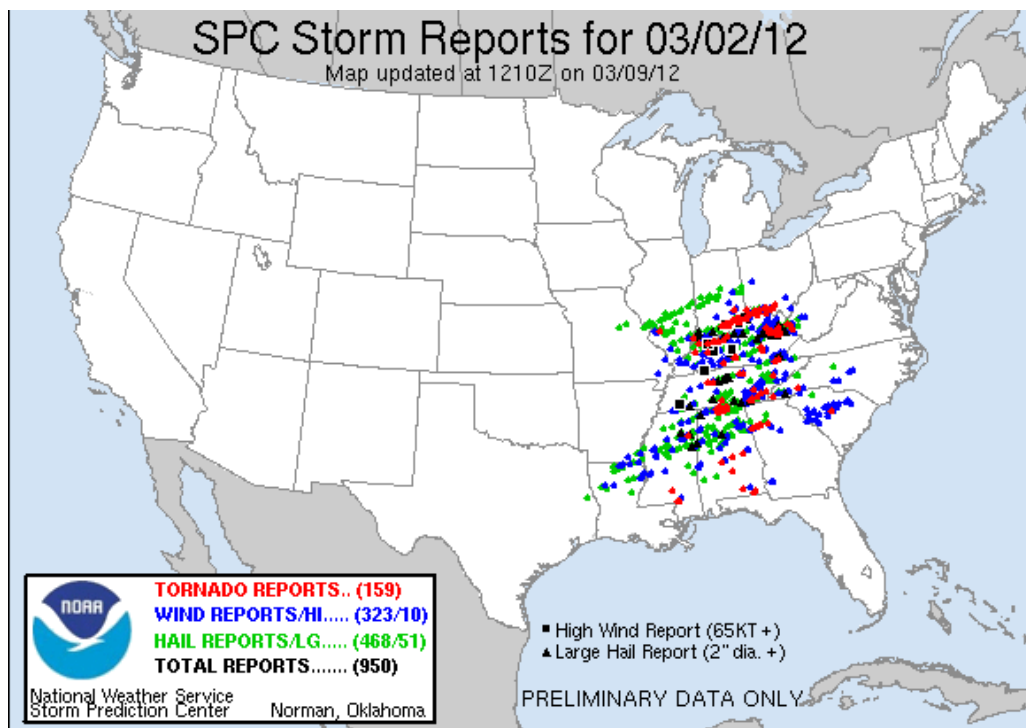


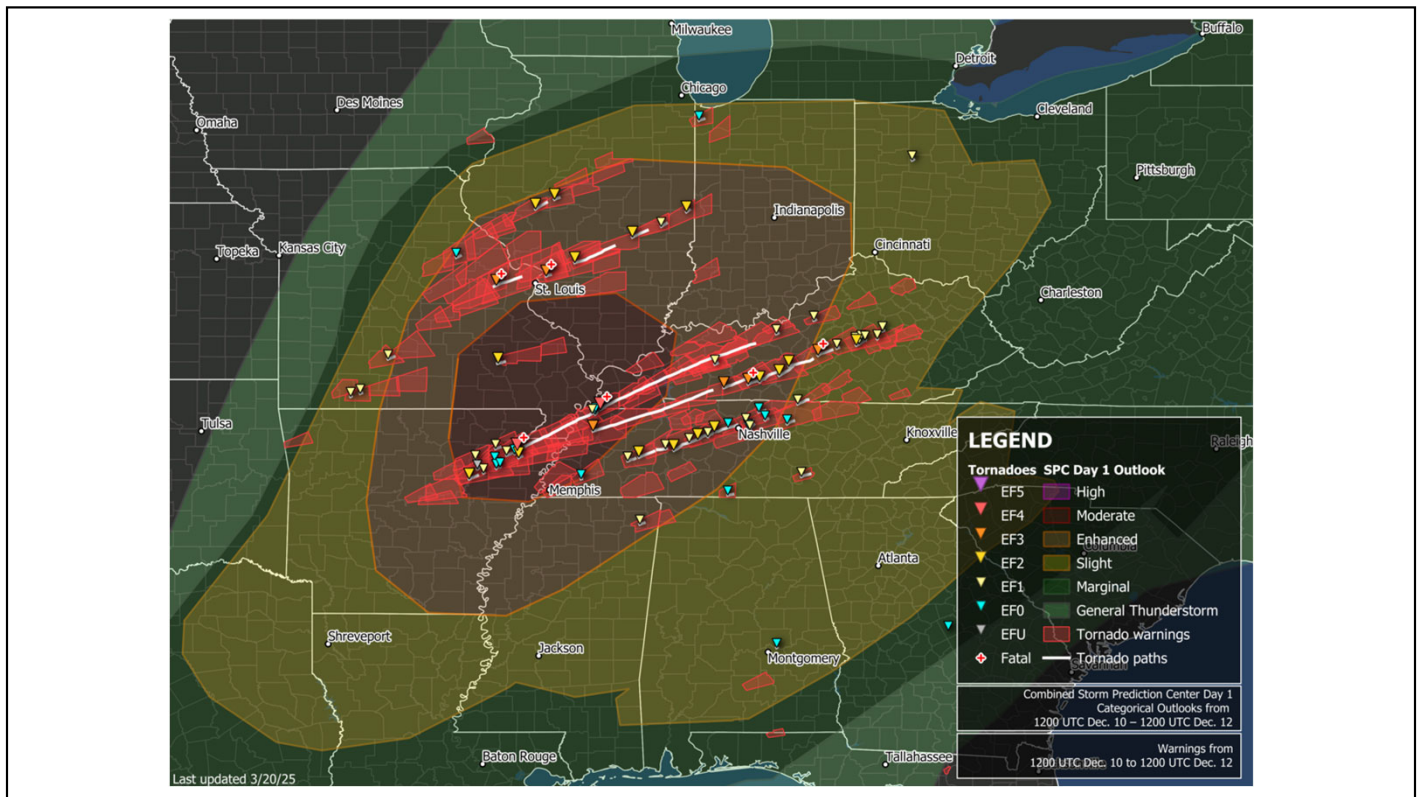
Figure 1. The expanding bull's-eye effect illustrated with the 2013 Moore, Oklahoma EF5 tornado footprint overlaid atop A) a theoretical metropolitan region and B) Wichita, Kansas, for the years of 1950 through 2100.

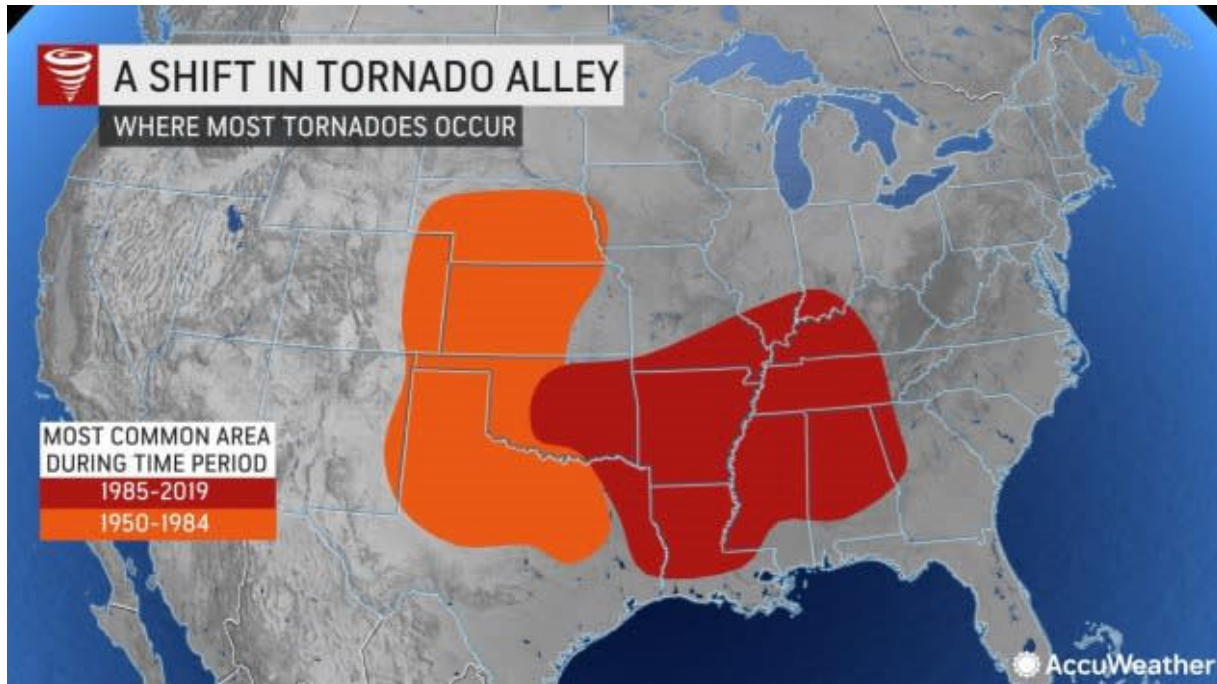
Tornado Archive Data Explorer



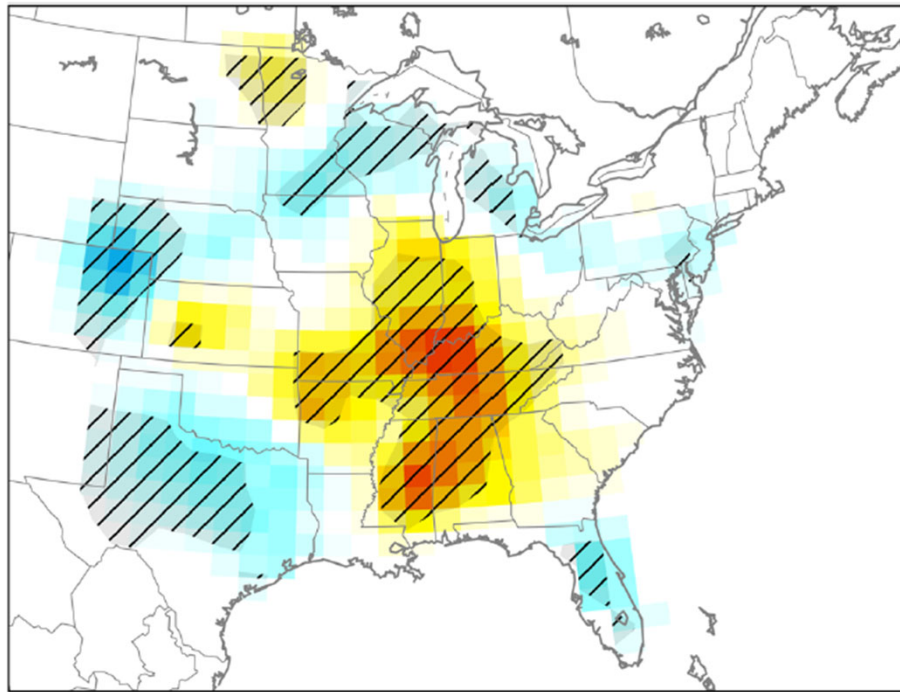


The storms resulted in 41 tornado-related fatalities, 22 of which occurred in Kentucky. 70 tornadoes





Both locations offer a mountain free zone for continental polar air masses from the north to collide with marine tropical airmass from the south. The location of this tornado ‘alley’ has shifted in the last few decades toward the southeastern US



Blue areas indicate where tornadoes have decreased since 1979. Yellow and red areas are where they have increased. Hatched lines indicate statistically significant changes.

doi:10.1038/s41612-018-0048-2

Severe thunderstorms accompanied by tornadoes, hail, and damaging winds cause an average of 5.4 billion dollars of damage each year across the United States, and 10 billion-dollar events are no longer uncommon. This overall economic and casualty risk—with over 600 severe thunderstorm related deaths in 2011—has prompted public and

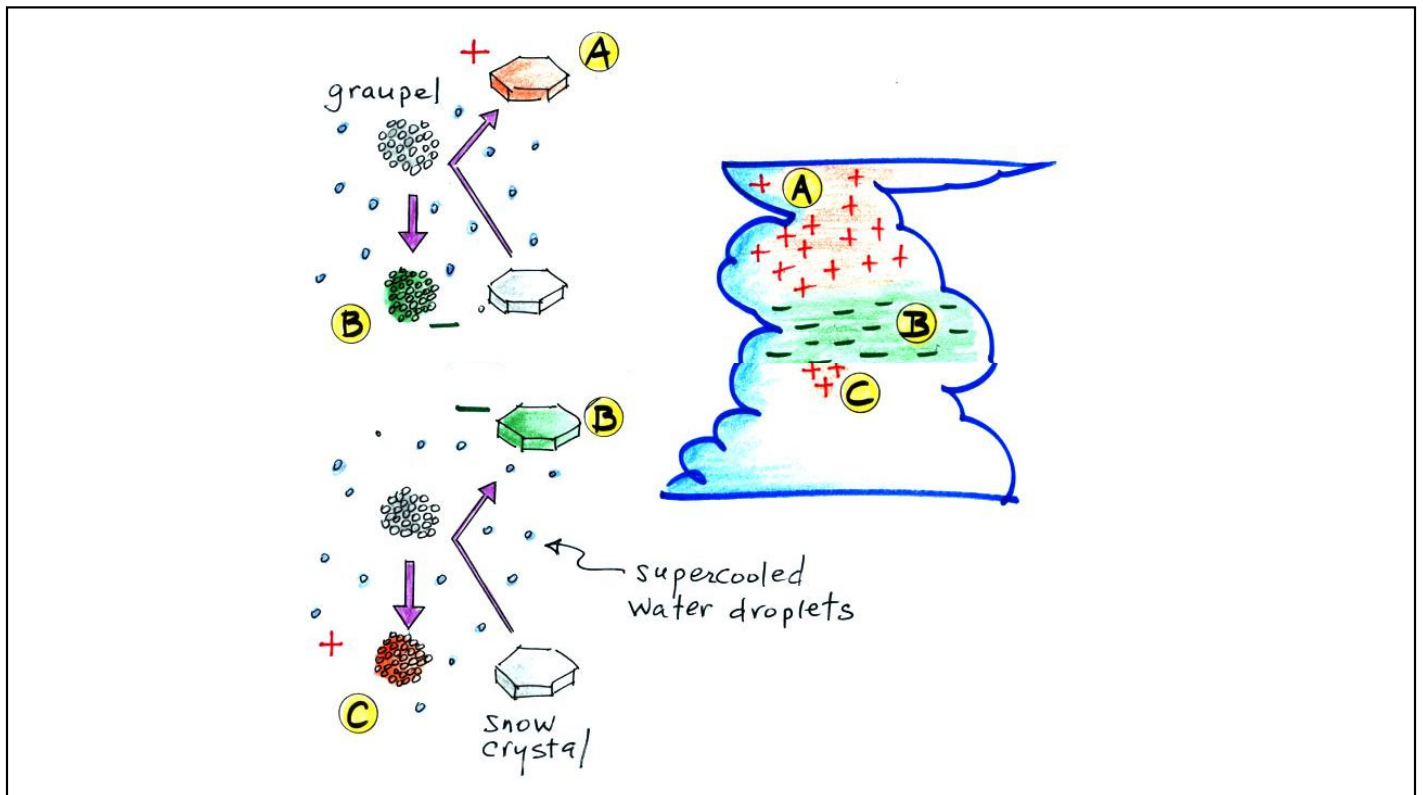
scientific inquiries about the impact of climate change on tornadoes. We show that national annual frequencies of tornado reports have remained relatively constant, but significant spatially-varying temporal trends in tornado frequency have occurred since 1979. Negative tendencies of tornado occurrence have been noted in portions of the central and southern Great Plains, while robust positive trends have been documented in portions of the Midwest and Southeast United States.

This temporal change of spatial patterns in tornado activity for successive cold and warm periods may be suggestive of climate change effects yet warrants the climatological study of meteorological parameters responsible for tornado formation



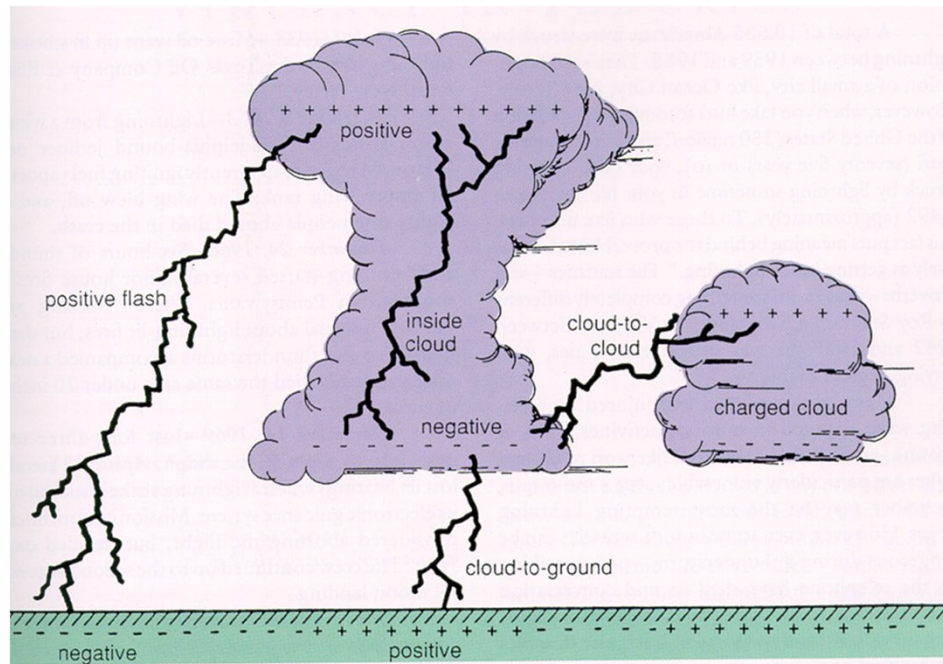


<https://map.blitzortung.org/>



The non-inductive charge separation process

http://www.atmo.arizona.edu/students/courselinks/spring13/atmo589/ATMO489_online/lecture_11/lect11_cloud_electrification.html



Electrons flow from the negative to the positive



<https://youtu.be/NUpN1PsgTOU?si=aUQB2IRlkBcsj6IX>

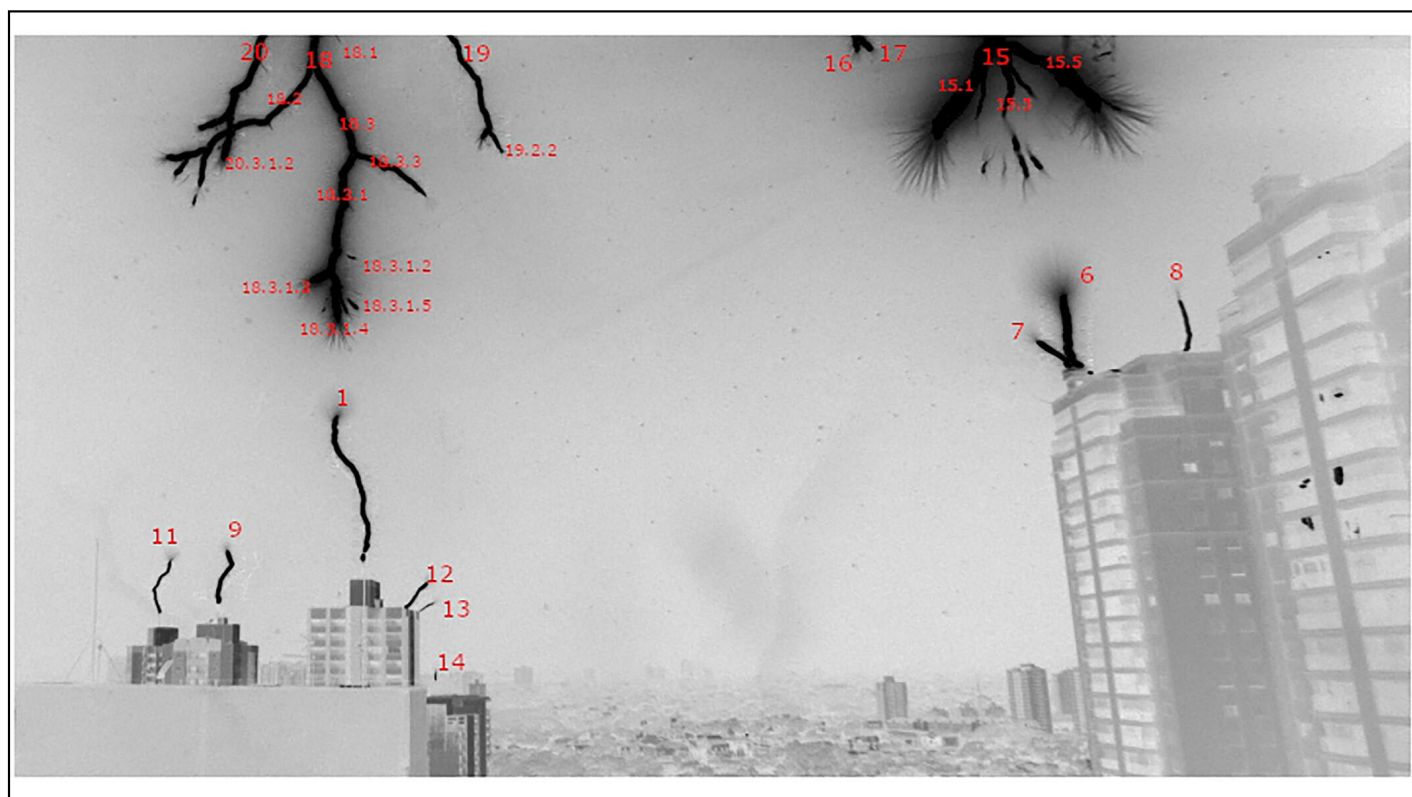


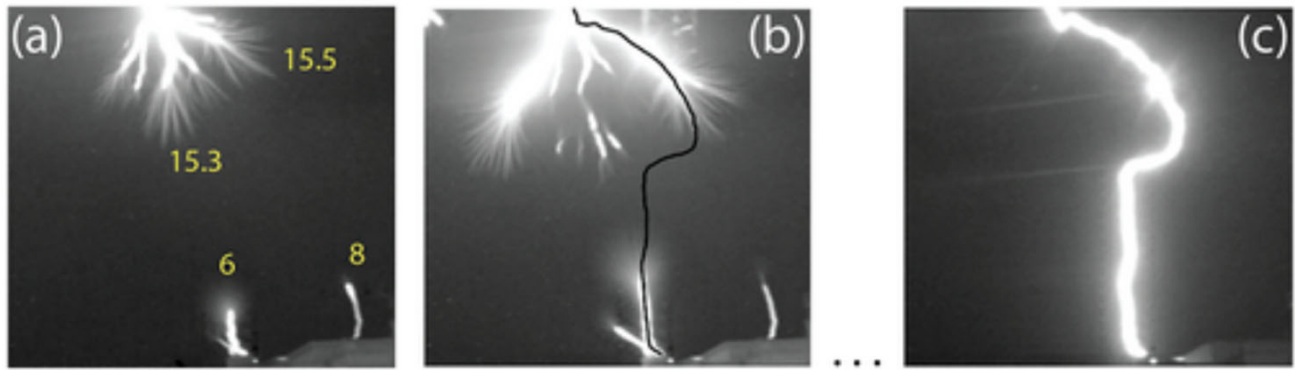
Figure 2. Selected frames from a Phantom high speed camera at 7200 frames per second showing a network of negative stepped leaders approaching a hilltop as an upward leader from a radio tower prepares to attach, completing the circuit with the upward return stroke.

Lightning rods initiate upward leader and hence attempt to direct the lightning rod.

<http://hyperphysics.phy-astr.gsu.edu/hbase/electric/ligseq.html#c2>

<http://www.ztresearch.com/>







<https://youtu.be/vbQtqsolWjs?si=4BOLHxSC85l7QxrC>